

COVER PAGE

Analysis of the 2017 Equifax Data Breach

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1. Executive Summary

This project analyzes the 2017 Equifax data breach, one of the largest cybersecurity breaches in history. Equifax, a major credit reporting agency, suffered a breach that exposed the personal information of about 147 million people. The breach occurred because attackers exploited an unpatched Apache Struts vulnerability known as CVE-2017-5638. The incident showed the importance of patch management, vulnerability scanning, network monitoring, encryption, and incident response.

2. Scope of the Project

The scope of this project is to analyze:

- The target organization affected by the breach
- The attack vector used by the attackers
- The vulnerability exploited
- The timeline of the breach
- The assets affected
- The impact on customers and the organization
- The incident response process
- Security controls that could have prevented or reduced the impact

3. Target Organization

Organization: Equifax Inc.

Industry: Credit Reporting / Financial Services

Breach Year: 2017

Affected Users: Approximately 147 million people

Type of Breach: Web application compromise and data exfiltration

Equifax stores highly sensitive consumer data such as names, Social Security numbers, birth dates, addresses, and driver's license numbers, making it a high-value target for cybercriminals

and nation-state attackers. The FTC states that Equifax announced in September 2017 that the breach exposed the personal information of 147 million people. ([Federal Trade Commission](#))

4. Assets, Threats, and Vulnerabilities

Category	Details
Assets	Consumer personal information, credit records, web applications, databases, internal systems
Threats	Data theft, identity theft, web application exploitation, unauthorized access, regulatory penalties
Vulnerabilities	Unpatched Apache Struts vulnerability, weak vulnerability management, poor detection capability, certificate monitoring failure, inadequate segmentation

5. Attack Vector and Initial Compromise

The attackers exploited a known vulnerability in Apache Struts, identified as **CVE-2017-5638**. Apache had released a patch in March 2017, but Equifax failed to apply the patch to the affected web application. The U.S. House Oversight report described the breach as preventable and noted that the vulnerability was publicly disclosed on March 7, 2017, with DHS alerting Equifax the next day. ([House Oversight Committee](#))

Once the attackers gained access, they were able to move through Equifax systems and access sensitive consumer information. The breach went undetected for weeks, partly because encrypted traffic inspection was not working properly due to an expired digital certificate, according to GAO reporting and related analysis. ([GAO](#))

6. Timeline of the Breach

Date / Period	Event
March 7, 2017	Apache Struts vulnerability publicly disclosed and patch released
March 8, 2017	DHS alerted Equifax about the vulnerability
May 2017	Attackers exploited the unpatched Apache Struts vulnerability
May–July 2017	Attackers accessed and exfiltrated sensitive data
July 2017	Equifax detected suspicious activity
September 2017	Equifax publicly disclosed the breach
2019	Equifax agreed to a global settlement with FTC, CFPB, and states

7. Impact of the Breach

Data Loss

The breach exposed highly sensitive personal information including:

- Names
- Social Security numbers
- Birth dates
- Addresses
- Driver's license numbers

This type of data is extremely dangerous because it can be used for identity theft and financial fraud.

Financial Impact

Equifax agreed to a global settlement of at least **\$575 million**, potentially up to **\$700 million**, with the FTC, CFPB, and U.S. states and territories. ([Federal Trade Commission](#))

Regulatory and Legal Impact

The breach led to investigations, lawsuits, regulatory penalties, and public criticism of Equifax's cybersecurity practices. The CFPB also announced the settlement with Equifax, stating that it involved up to \$700 million in monetary relief. ([Consumer Financial Protection Bureau](#))

Reputation Impact

The breach damaged public trust because Equifax's business depends on protecting consumer credit information. The incident became a major example of how failure to patch known vulnerabilities can create serious consequences.

8. Incident Response Review

Equifax eventually detected the breach and shut down the exploit, but the response was criticized because:

- The vulnerability was known before the attack
- The patch was available but not applied properly
- Detection was delayed
- Public disclosure happened weeks after discovery
- The breach affected a very large number of people

The incident showed that incident response is not only about reacting after an attack, but also about preparation, monitoring, communication, and rapid remediation.

9. Security Controls That Could Have Prevented or Reduced the Breach

Security Control	How It Would Help
Patch Management	Apply Apache Struts patch quickly after disclosure
Vulnerability Scanning	Detect outdated or vulnerable software
Asset Inventory	Identify all systems running Apache Struts
Web Application Firewall	Block exploitation attempts
Network Segmentation	Limit attacker movement after initial access
SIEM Monitoring	Detect suspicious activity earlier
Certificate Monitoring	Prevent failure of encrypted traffic inspection
Data Encryption	Reduce exposure if data is stolen
Least Privilege	Limit access to sensitive databases
Incident Response Plan	Improve detection, containment, communication, and recovery

10. Evidence and Screenshots

Overview

The screenshots included in this section serve as supporting evidence for the research, analysis, and documentation conducted during this cybersecurity case study. The screenshots demonstrate the research process, vulnerability analysis, timeline development, risk assessment activities, and security control recommendations related to the Equifax Data Breach.

Screenshot 1: FTC Equifax Data Breach Settlement

File Name: equifax_ftc_settlement_page.png

Description

This screenshot shows the Federal Trade Commission (FTC) official page regarding the Equifax Data Breach Settlement. The page provides information about the breach, the number of affected individuals, consumer compensation, and regulatory actions taken against Equifax.

Purpose

The screenshot serves as evidence that the breach resulted in significant legal and financial consequences and demonstrates the regulatory response to the incident.

Key Findings

- Approximately 147 million individuals were affected.
- Equifax faced substantial financial penalties.
- The breach triggered multiple investigations and lawsuits.
- The incident became one of the largest consumer data breaches in history.

The screenshot shows the FTC website with the following elements:

- Header:** "An official website of the United States government" with a link "Here's how you know". Navigation links include "Español", "Report Fraud", "Submit Public Comments", and "Search the Legal Library".
- FTC Logo and Name:** "FEDERAL TRADE COMMISSION" and "PROTECTING AMERICA'S CONSUMERS".
- Menu:** "Enforcement", "Policy", "Advice and Guidance", "News and Events", "About the FTC", and a search icon.
- Breadcrumbs:** "Home / Enforcement / FTC Refund Programs". A link "Vea esta página en español" is on the right.
- Title:** "Equifax Data Breach Settlement".
- Date and Social Media:** "November 2024" with icons for Facebook, X, and LinkedIn.
- Section Header:** "Important Settlement Update".
- Text:** "The settlement administrator has been sending out payments for out-of-pocket losses, time spent claims, and other cash benefits. In November 2024, the settlement administrator will be sending prepaid cards with additional payments to those who".
- Related Cases:** A section titled "Related Cases" with a link "Equifax, Inc."

Screenshot 2: Apache Struts Vulnerability Research

File Name: apache_struts_vulnerability_research.png

Description

This screenshot displays information regarding the Apache Struts vulnerability (CVE-2017-5638), which was the primary attack vector exploited during the Equifax breach.

Purpose

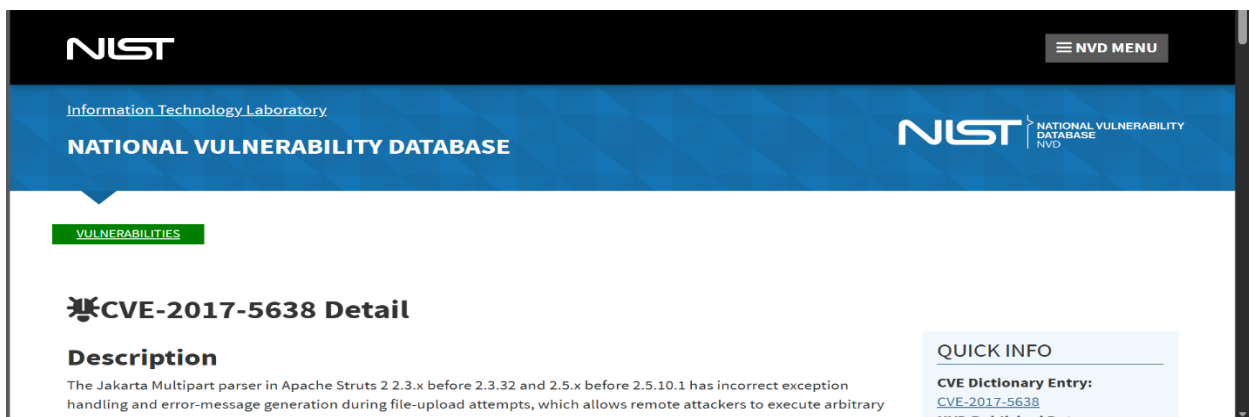
The screenshot provides evidence of the technical vulnerability that allowed attackers to gain unauthorized access to Equifax systems.

Key Findings

- Vulnerability Identifier: CVE-2017-5638
- Affected Technology: Apache Struts Framework
- Severity: Critical
- Patch Available Prior to Attack
- Exploited Due to Failure to Apply Security Updates

Security Lesson

Organizations must maintain effective patch management processes to prevent exploitation of known vulnerabilities.



The screenshot shows the NIST National Vulnerability Database (NVD) interface. At the top, the NIST logo is on the left, and a "NVD MENU" button is on the right. Below the header, the text "Information Technology Laboratory" and "NATIONAL VULNERABILITY DATABASE" are displayed. A green "VULNERABILITIES" tab is selected. The main content area is titled "CVE-2017-5638 Detail". Under the "Description" section, it states: "The Jakarta Multipart parser in Apache Struts 2.2.3.x before 2.3.32 and 2.5.x before 2.5.10.1 has incorrect exception handling and error-message generation during file-upload attempts, which allows remote attackers to execute arbitrary". To the right, a "QUICK INFO" box contains the "CVE Dictionary Entry: CVE-2017-5638" and a partially visible "NVD Published Date:" field.

Screenshot 3: NIST / CISA Security Guidance

File Name: nist_or_cisa_patch_management_reference.png

Description

This screenshot shows cybersecurity guidance from either the National Institute of Standards and Technology (NIST) or the Cybersecurity and Infrastructure Security Agency (CISA).

Purpose

The screenshot demonstrates industry best practices for vulnerability management, risk assessment, patch management, and cybersecurity governance. Key Findings

- Importance of continuous monitoring.
- Need for timely patch deployment.
- Security frameworks improve organizational resilience.
- Risk management is essential for protecting critical assets.

Security Lesson

Following recognized security frameworks significantly reduces organizational exposure to cyber threats.



Screenshot 4: Cybersecurity Timeline Design

File Name: timeline_design.png

Description

This screenshot shows the development of the historical cybersecurity timeline created using Draw.io or another visualization tool.

Purpose

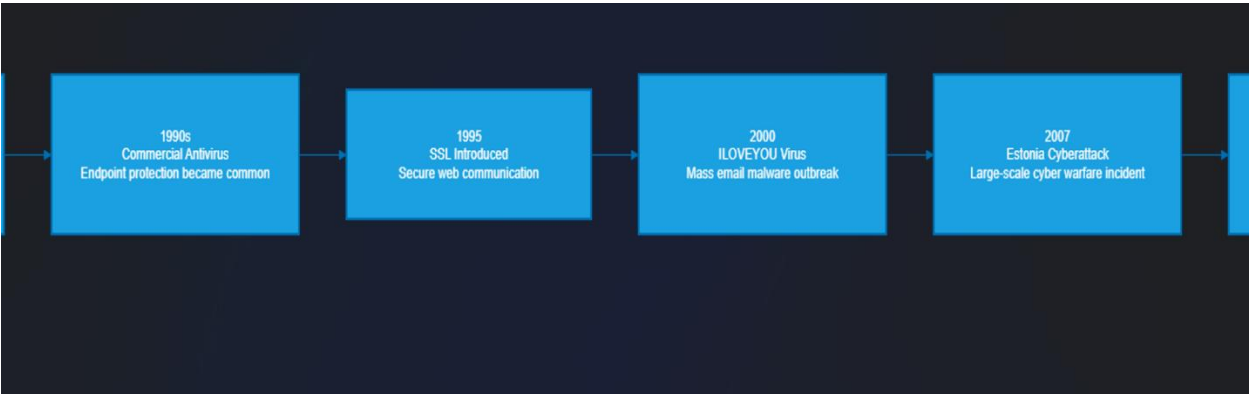
The screenshot demonstrates the research and visualization process used to organize cybersecurity milestones from 1971 to the present.

Timeline Milestones Included

- Creeper Program (1971)
- Reaper (1972)
- Morris Worm (1988)
- ILOVEYOU Virus (2000)
- Stuxnet (2010)
- WannaCry (2017)
- SolarWinds (2020)
- Zero Trust Architecture (2020)
- AI-Driven Security (2023-Present)

Security Lesson

Cybersecurity continuously evolves in response to emerging threats and technological advancements.



Screenshot 5: Risk Analysis Table

File Name: risk_analysis_table.png

Description

This screenshot presents the risk analysis conducted as part of the Equifax breach case study.

Purpose

The risk analysis identifies critical assets, threats, and vulnerabilities associated with the incident.

Example Risk Assessment

Asset	Threat	Vulnerability
Customer Data	Data Theft	Unpatched Software
Web Application	Unauthorized Access	Apache Struts Vulnerability
Database Systems	Data Exfiltration	Weak Monitoring Controls

Security Lesson

Risk assessments help organizations identify weaknesses before attackers can exploit them.

ASSET	THREAT	VULNERABILITY
Customer Data	Data Theft	Unpatched Apache Struts
Web Application	Unauthorized Access	Vulnerability Exploitation
Database	Data Exfiltration	Weak Monitoring

Screenshot 6: Security Controls and Mitigation Table

File Name: security_controls_table.png

Description

This screenshot shows the recommended security controls that could have prevented or reduced the impact of the Equifax breach.

Purpose

The screenshot demonstrates the relationship between identified vulnerabilities and appropriate security controls.

Recommended Controls

Security Control	Purpose
Patch Management	Address known vulnerabilities
Vulnerability Scanning	Detect security weaknesses
SIEM Monitoring	Identify suspicious activity
Web Application Firewall	Block malicious traffic
Multi-Factor Authentication	Strengthen access control
Network Segmentation	Limit attacker movement
Incident Response Plan	Improve breach response

Security lesson

Layered security controls provide stronger protection than relying on a single security mechanism.



CONTROL	PURPOSE
Patch Management	Fix vulnerabilities
SIEM	Detect attacks
WAF	Block web attacks
MFA	Improve authentication
Network Segmentation	Limit attacker movement
Vulnerability Scanning	Find weaknesses



Summary of Evidence

The screenshots included in this report support the technical analysis of the Equifax Data Breach and demonstrate the research process used to identify attack vectors, vulnerabilities, business impacts, and recommended security controls. Together, they provide evidence of the methodology used in this cybersecurity case study and reinforce the importance of proactive security management, vulnerability remediation, and continuous monitoring.

11. Methodology and Tools Used

Tool / Method	Purpose
Web Research	Gather reliable information about the breach
Timeline Analysis	Understand breach progression
Risk Assessment	Analyze threats, vulnerabilities, and impact
Control Mapping	Recommend preventive controls
Draw.io / Word	Create timeline and report documentation

12. Lessons Learned

The Equifax breach teaches that cybersecurity failures often happen because of poor basic security practices. The attack did not require a completely unknown vulnerability; it exploited a known issue for which a patch already existed. This shows the importance of patch management, asset inventory, monitoring, and accountability.

The breach also shows that organizations handling sensitive data must treat cybersecurity as a business-critical function, not only an IT responsibility.

13. Conclusion

The 2017 Equifax data breach was a major cybersecurity failure caused mainly by poor vulnerability management and delayed detection. Attackers exploited an unpatched Apache Struts vulnerability and gained access to highly sensitive consumer data. The breach affected millions of people and resulted in major financial, legal, and reputational consequences.

To prevent similar attacks, organizations must implement strong vulnerability management, continuous monitoring, encryption, network segmentation, and effective incident response planning. The Equifax breach remains an important lesson in the need for proactive cybersecurity governance and disciplined security operations.

14. References

1. Federal Trade Commission. *Equifax Data Breach Settlement*. ([Federal Trade Commission](#))
2. U.S. House Oversight Committee. *The Equifax Data Breach Report*. ([House Oversight Committee](#))
3. Government Accountability Office. *Actions Taken by Equifax and Federal Agencies in Response to the 2017 Breach*. ([GAO](#))
4. Consumer Financial Protection Bureau. *CFPB, FTC and States Announce Settlement with Equifax*. ([Consumer Financial Protection Bureau](#))
5. Federal Trade Commission. *Equifax Inc. Case Summary*. ([Federal Trade Commission](#))

